TP Link Kasa and Tapo Platform Driver

Version 2.1.005.0000

Driver

This guide explains how to install and use Neil Colvin's TP-Link Kasa and Tapo platform driver for Crestron Home. One platform instance discovers supported devices on the local network and publishes separate lighting, outlet, sensor and button children. Assign the required children to rooms in Crestron Home configuration.

Driver version 2.1.005.0000. Package and help filename: NeilColvin_Platform_KasaTapo_IP_V2. The driver uses KasaTapoClient 2.0.1 for local device communication.

Notes and Recommendations

Set up each physical device in its manufacturer's app first. Pair hub sensors and buttons with their hub before discovering them in Crestron Home. Keep the processor and devices on a network that permits local discovery and direct communication; network isolation can prevent discovery even when the manufacturer's app still works.

The driver does not discover or control devices through the TP-Link cloud. Tapo credentials authenticate local connections to devices that require them. This is separate from the initial setup performed in the manufacturer's app.

Use one platform instance. Select only the children required for the installation; discovery can include other supported devices on the same network. Changes to a room tile's name do not necessarily change the physical device's name shown on its detail page.

System Requirements and Dependencies

- A supported Crestron Home 4-Series processor and a compatible Crestron Home app/configuration tool. Exact versions exercised for this driver are listed in the test environment section.
- Supported Kasa or Tapo devices reachable locally from the processor. Hub-connected devices require a reachable, configured hub.
- Tapo user name and password for devices requiring authenticated local access. Leave both blank for a Kasa-only installation.
- No separate .NET library installation is required on the processor; the driver package contains its required application dependencies.

Installation/Upgrade Instructions

Make the driver package available to the processor, then add the TP-Link KasaTapoPlatform platform driver using Crestron Home configuration. Enter the optional Tapo credentials and complete the platform settings. Wait for local discovery to populate the platform's available children.

Assign each required child to its room and complete the child's configuration. For a plug or strip socket, Treat As Light selects a lighting load instead of the outlet interface. Configure each socket separately. Do not treat a socket as a dimmer unless the actual device advertises dimming support.

Enable Light Polling when changes made outside Crestron Home must be reflected automatically. Light Poll Interval controls that polling cadence. Sensor/Button Poll Interval is a fallback until a hub child reports its own cadence; hub polling then uses half the shortest reported interval among its connected children.

Sensor and button children provide Report Interval in seconds. Zero preserves the device's existing interval. A nonzero value changes the physical device's reporting interval where supported. Button children also offer Allow Double Click; configure it only when the device supports that feature.

The optional Processor Baseline Workaround is disabled by default. It uses processor SSH access to address a documented lighting tuning-mode issue. It requires unique light names and processor SSH credentials. Leave it disabled unless the installation needs the workaround; see the repository's Known Issue section for its current scope.

End-User Experience

Lights

Lights use Crestron Home's native lighting interface. On/off, brightness, color and color temperature are exposed according to the physical device's advertised capabilities. An on/off-only device does not gain dimming or color control from this driver.

Outlets

Outlet tiles appear in the assigned room and provide on/off control. A basic outlet has no separate detail page. An outlet with energy reporting provides a detail page with a Power toggle and the telemetry reported by that device, such as power, voltage, current and energy totals. Unsupported measurements are not a promise of capability.

Sensors

Sensor detail pages are read-only. Depending on the sensor, they show temperature, humidity, contact, motion or leak state and available battery information. These values depend on the sensor's reporting interval and hub communication; opening the page does not force a physical measurement.

Buttons

Button tiles and detail pages show the most recent reported gesture, its time and available battery information. Physical button gestures can be used with the events exposed to Crestron Home programming. The detail page displays telemetry and does not remotely press the physical button.

Outlet feedback and the 2.1.5 update

Driver 2.1.005.0000 | GitHub/NuGet 2.1.5 | 2 November 2026

Confirmed outlet state

The outlet state property, status and icon follow successful device readback. Outlet Turned On and Outlet Turned Off are raised once per confirmed transition, including changes detected after using the Kasa/Tapo app or physical control. A failed command or readback does not confirm a new state. Repeated unchanged polls and commands do not repeat transition events.

The first successful read establishes initial state without raising an on/off event. Queued toggles use the preceding command's readback, and retries keep the same requested target.

Polling for changes outside Home

Enable Enable Light Polling in the platform settings. Plain outlets also use Light Poll Interval (Seconds); the default is 15 seconds. External feedback follows a successful poll, so allow the configured interval and device response time.

Third-party toggle tiles

The driver's own outlet UI uses its state directly. Other tile drivers need feedback programming using their own state/display actions, where supported:

1. Outlet Turned On: set the tile's displayed feedback to ON.
2. Outlet Turned Off: set the tile's displayed feedback to OFF.

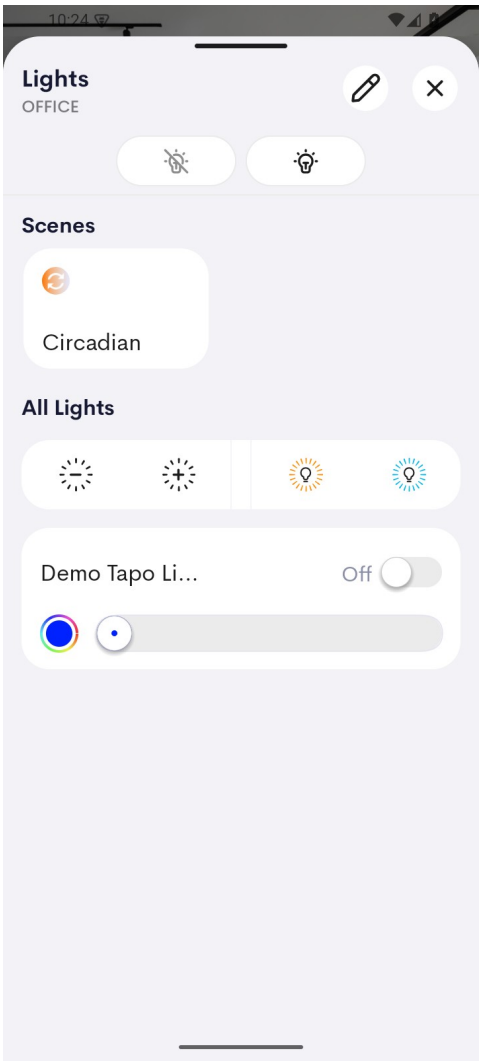
Use feedback actions that do not send another command to the plug. Action names and startup synchronization depend on the tile driver. Publishing outletIsOn does not automatically connect unrelated tiles. After setting up the feedback, switch the plug off and on to check both directions.

Upgrade and validation

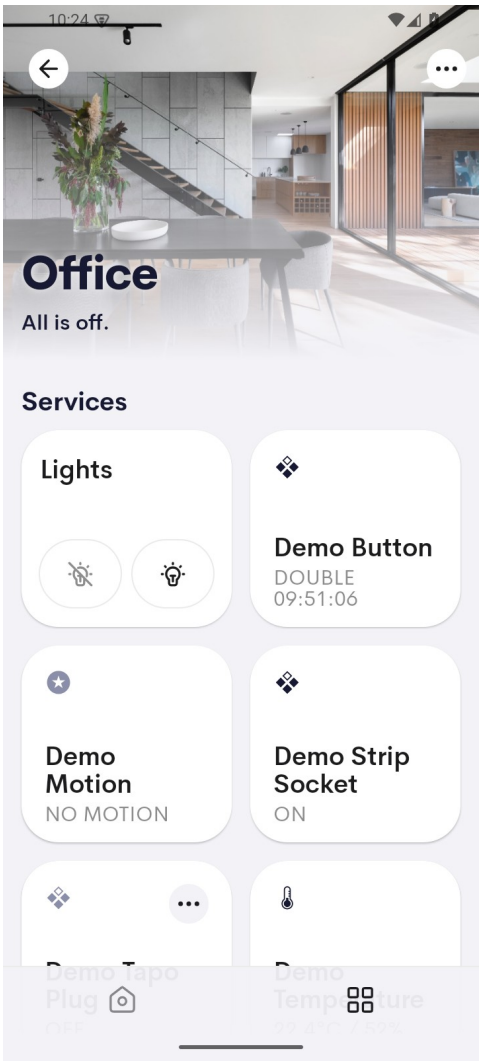
Import the package, then update the existing KasaTapoPlatform driver in Home Setup or Configure Pro. Importing alone does not update the running driver. Removing and re-adding configured children is not a routine update step. Existing device identities, programmable IDs and light/dimmer paths are retained.

A customer reported the equivalent 2.1.004.0001 test fix working on a CP4-R with 10-second polling and both Ultamation and Control Programming LLC toggle tiles. Home control and feedback for Kasa-app changes were confirmed. The plug model and a physical-button test were not specified; this report is separate from automated SDK/processor tests.

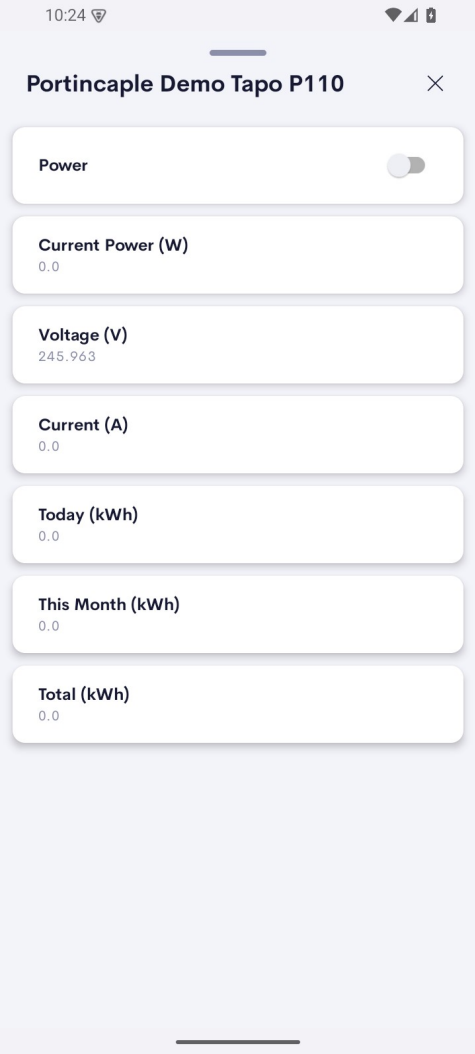
Support: contact form. No GitHub account is required.



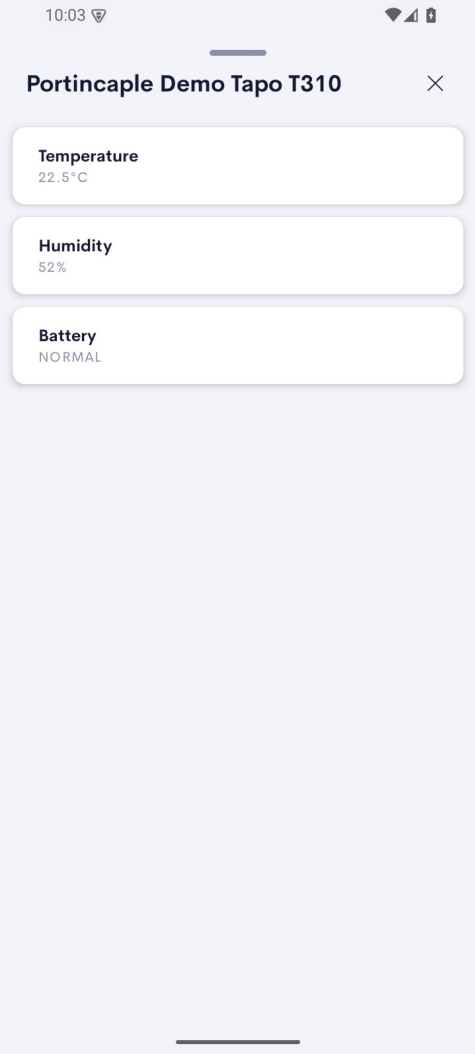
Native Lights view with a Demo bulb. Available controls follow the physical device capabilities. Use the individual load controls; room-wide controls affect every light in the room.



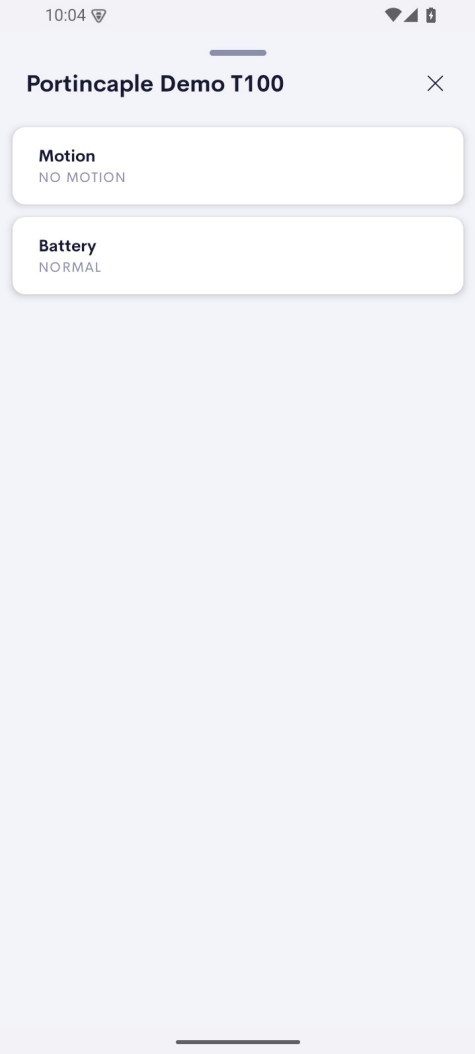
Room view with outlet, sensor and button tiles. A basic outlet toggles from its tile. The three-dot menu on an energy-capable outlet opens its detail page.



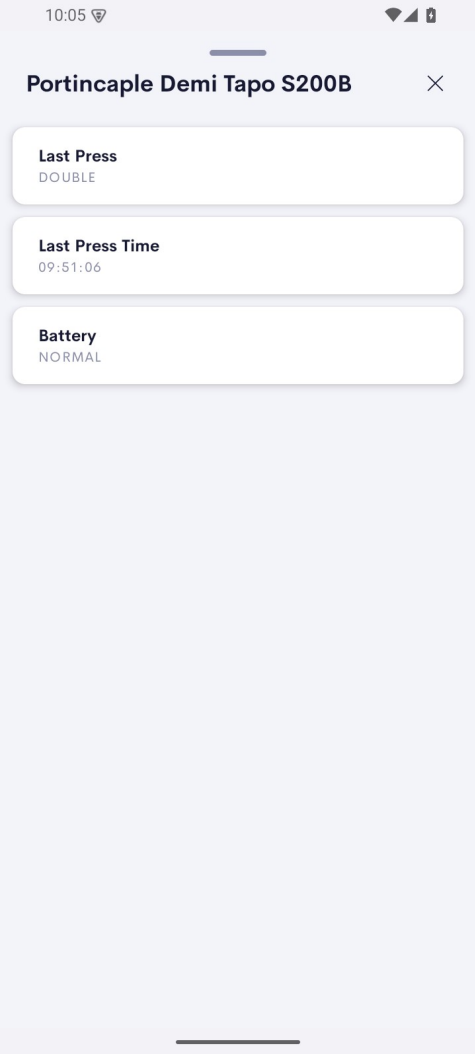
Energy-capable outlet detail page. Power controls this outlet; the remaining rows show its reported telemetry.



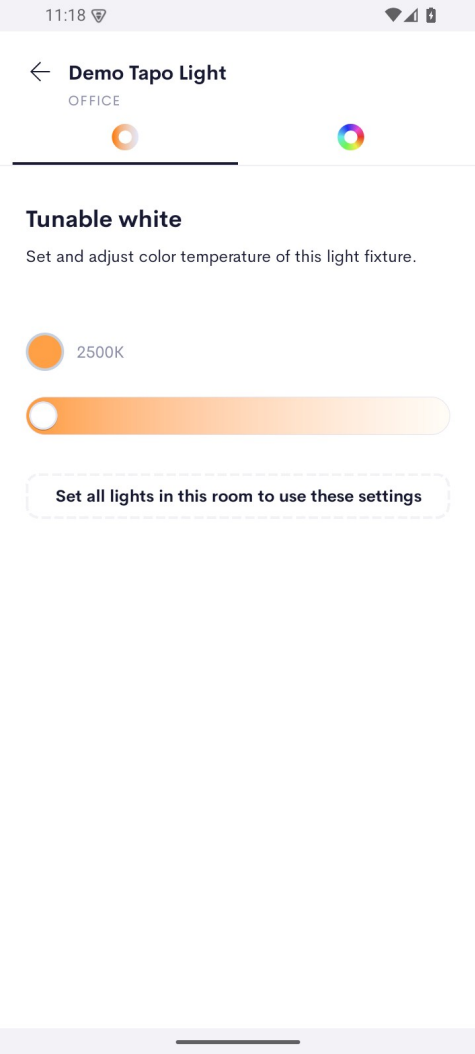
Temperature and humidity sensor page with battery status. Values update according to the sensor and hub reporting cadence.



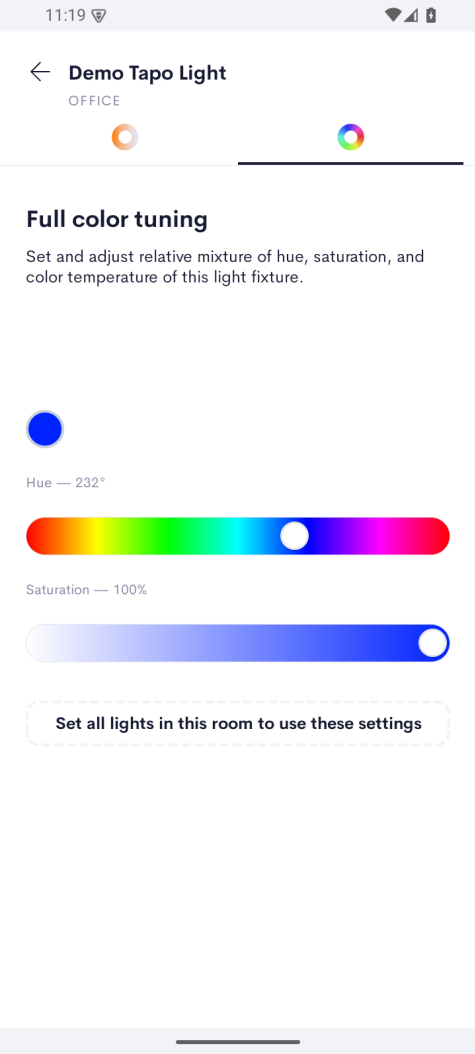
Motion sensor page. Motion and battery information are read-only.



Button page showing the most recently reported gesture, time and battery state. This page does not remotely press the button.



Native tunable-white page for an individual color-capable bulb. The room-wide apply button affects other lights and is separate from changing this bulb.



Native full-color page with individual hue and saturation controls. The room-wide apply button affects every light in the room.

Limitations/Known Issues

- Device capabilities vary by model, hardware revision and firmware. Discovery alone does not establish that every control or measurement is supported or tested.
- Cloud-only devices, remote cloud discovery and devices isolated from local network discovery are outside this driver's scope.
- Smart wall switches are currently exposed as lighting loads; the plug/strip Treat As Light option does not imply an equivalent plain-outlet mode for every wall switch.
- External state feedback requires the applicable polling/reporting path. An online device and a displayed last reading do not prove that a new sensor measurement has occurred.
- A failed network or authenticated connection can delay discovery, control or feedback. Check device reachability, credentials and the manufacturer's app when troubleshooting.
- On affected Crestron Home firmware, the native tuning page can initially show White controls while the bulb is in Color mode, or the reverse. A brief wrong-color transition can also occur at power-on.

The optional Processor Baseline Workaround is disabled by default; consult the repository for its scope and configuration.

Supported Features

- Local discovery and managed children for supported lighting, outlets, hub sensors and buttons.
- Native lighting capabilities, independent strip-socket control and optional energy telemetry.
- Cached managed-device metadata and background child connections for driver startup and reload.
- Per-child capabilities and programmable properties/events where supported by the physical device.

Test Environment

Help illustrations and compatibility checks used a CP4-R running Crestron Home 4.011.0322 and the Crestron Home Android app 4.11.1+pr. Screens were captured using driver 2.1.1; the illustrated controls are unchanged in 2.1.5.

- Tapo P110 hardware 1.0, firmware 1.4.8 Build 260804 Rel.045837; Tapo L530 hardware 3.0, firmware 1.4.4 Build 260716 Rel.091150.
- Kasa KP303(UK) hardware 1.0, firmware 1.0.5 Build 210428 Rel.144721; Tapo H100 hardware 1.0, firmware 1.7.10 Build 260725 Rel.042335.

The H100 setup included T310 temperature/humidity, T100 motion and S200B button children. Individual child hardware and firmware versions were not returned by the inspected system-information response and are not claimed here.

Supported Models

Observed models include P110(UK), KP303(UK), L530E(EU), H100(UK), T310, T100 and S200B. Compatibility depends on model, hardware revision and firmware. The driver discovers additional supported KasaTapoClient models according to their reported capabilities; consult the repository for current scope and limitations.

Contact Information

Developer: Neil Colvin.

Support form: https://oznetmaster.github.io/support/ . No GitHub account is required. Include the driver version, device model and firmware, processor/app versions, symptoms and steps to reproduce. Do not include passwords or credentials.

Source, documentation and release history: https://github.com/oznetmaster/KasaTapoCrestronDriver .

Version History

2.1.2 adds packaged user help, explicit support contact information and a developer-prefixed package filename. See the repository changelog for earlier driver behavior changes.

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The package includes THIRD-PARTY-NOTICES.txt with the driver license and the reviewed licenses and notices for its merged dependencies, including KasaTapoClient, Bouncy Castle, SSH.NET, Hafner compatibility attributes and Microsoft .NET libraries. Those components retain their own license terms.